

The Earliest Inequality: Two-Parent Advantage, Food Inflation, and Birth Outcomes in the Brazilian Semiarid¹

Abstract: This paper estimates the effect of exposure to food inflation during pregnancy on birth outcomes and documents how these outcomes vary by family structure. We combine administrative records on seven million births in the Brazilian Semiarid between 2000 and 2023 with plausibly exogenous variation in local food inflation. Higher food inflation during the third trimester of pregnancy reduces birthweight by 3 to 8 grams for every 10-percentage-point increase in food prices. However, exposure during the first and second trimesters has no clear effect. Regarding family structure, conditional on mother's age, education, race, child's sex, municipality fixed effects, and year fixed effects, children born into two-parent households are 19 grams heavier at birth than those born to single mothers. The two-parent advantage is considerably larger among Black and younger mothers, reaching 25 grams, and rises further to 26 grams among Black mothers with low levels of education. We also found that higher maternal education reduces the two-parent advantage, while legal marriage strengthens it. Both the negative impact of third-trimester food inflation and the two-parent advantage remain robust across a wide range of specifications and sample restrictions, including the exclusion of the COVID-19 period.

JEL: Q18, Q54, I15.

Keywords: Two-parent advantage, food inflation, birth outcomes, climate-vulnerable areas.

Statements and Declarations

Competing Interests: The authors declare no competing interests.

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1. Introduction

Children raised in two-parent households consistently outperform their peers across a broad range of health, educational, and socioeconomic outcomes (McLanahan and Sandefur, 1994; Waldfogel et al., 2010). These disparities arise early in life, including during the prenatal period (Kearney and Levine, 2017), and may have lasting consequences by contributing to poverty traps that constrain intergenerational mobility (Currie and Vogl, 2013). A key mechanism underlying differences in birth outcomes is the sensitivity of fetal development to maternal environmental conditions. Exposure to stressors such as climate shocks, conflict, and economic fluctuations can adversely affect neonatal health. Prior research documents the effects of extreme heat (Deschênes et al., 2009; Andalon et al., 2016; Chen et al., 2020), environmental disasters (Oliveira et al., 2020; Carrillo et al., 2020), and prenatal psychological stress induced by violence and terrorism (Camacho, 2008), among other factors.

In this paper, we examine the effects of food inflation during pregnancy and family structure on birth outcomes. We focus on the Brazilian Semi-arid, one of the country's poorest and most climate-vulnerable regions. To do so, we combine individual-level administrative birth records from 2000 to 2023, which include information on maternal marital status, with municipality-level measures of food inflation derived from the food and beverages component of the Consumer Price Index (IPCA), Brazil's official inflation index.

Food inflation has been linked to various consequences, including increases in the number of strikes (Bellemare, 2015) and political instability (Arezki and Bruckner, 2011), as well as a higher incidence of obesity, reflecting a shift in the typical family menu toward more energy-dense, inexpensive, and readily available foods (Conklin et al., 2019; Robinson, 2023). When examining maternal nutrition, Almond and Mazumder (2011) identified a correlation between fasting during Ramadan and a decrease in birthweight. The literature exploring the relationship between food inflation (or prices) and newborn health outcomes focuses on low- or middle-income countries. For example, studies by Woldemichael et al. (2022) and Kidane and Woldemichael (2020) were conducted in Ethiopia. The underlying premise is that individuals experiencing poverty are more vulnerable to food price fluctuations, which can lead to food insecurity and hunger. In high-income countries, while food access in terms of quantity may be

less affected, higher food prices can still lead to poorer outcomes by reducing nutritional quality and dietary diversity (Brinkman et al., 2010).

Our analysis focuses on the Brazilian Semi-arid Region, which exhibits lower levels of socioeconomic development than the national average and heightened vulnerability to food insecurity. These conditions reflect widespread poverty and structural constraints on agricultural production arising from limited water availability and scarce natural resources (Sutherland et al., 1999; Fagundes et al., 2020; Figueiredo et al., 2021). Given these structural vulnerabilities, the premise suggests that infants in the Brazilian Semi-arid are particularly vulnerable to the effects of food inflation. These concerns are magnified by Brazil's recent experience of surging food inflation – between January 2020 and January 2023, the food category within the National Consumer Price Index (IPCA), as the Brazilian Institute of Geography and Statistics (IBGE) reported, increased by nearly forty percent (Brasil, 2025).

While food inflation and its effect on birthweight is the primary link we assess, our empirical design allows us to examine how vulnerability operates as an amplifier of this shock. We show that family structure² constitutes an additional layer of vulnerability: single mothers in an already fragile socioeconomic environment appear particularly exposed to food price shocks. The degree of endogeneity concern differs substantially between our two key variables. Food inflation is plausibly exogenous to individual households, as it is largely driven by aggregate supply conditions and market dynamics (Almond et al., 2011; Kidane and Woldemichael, 2020). While households are unlikely to precisely time fertility decisions in response to short-term price fluctuations, selection into births cannot be fully ruled out. Given the food price level before pregnancy, such fluctuations are typically unpredictable, surprising both economists and expectant mothers alike. As highlighted by Cruces et al. (2012) in their examination of economic crises in Argentina, the unpredictable nature of these macroeconomic variations leaves limited scope for a third variable to confound the estimated relationships with newborn health outcomes. Accordingly, we interpret food inflation as an exogenous shock, conditional on fixed effects and observed covariates.

The family structure, however, is inherently endogenous, as two-parent and single-parent families differ along several observable and unobservable characteristics. Children raised in

² Throughout the paper, we use the terms family structure and two-parent advantage interchangeably to denote the comparison between two-parent households and single-mother households.

two-parent households experience better health, educational, and socioeconomic outcomes than those raised by single mothers, largely reflecting differences in economic resources, parental time and its quality and family stability (Björklund et al., 2007; Waldfogel et al., 2010; Brown, 2004; Artis, 2007; Moschini, 2023). In our sample, for instance, mothers in two-parent households have slightly higher levels of education on average (3.60 years of schooling versus 3.40 years for single mothers). Maternal schooling proxies for income but imperfectly so. This is particularly relevant as marriage and consensual unions are more prevalent among wealthier households, suggesting that income itself may confound the relationship between family structure and birthweight.

Our empirical strategy involves three steps. First, we separate the food and beverages component of the inflation index. Second, we match this specific local food inflation index to the mother's city of residence and the child's month and year of birth. Third, we relate birth outcomes to trimester-specific exposure to municipality-level food inflation, assigning monthly food price changes to each stage of pregnancy based on the child's date of birth, while controlling for maternal characteristics (such as marital status and schooling), pregnancy characteristics (child's sex), as well as municipality and year fixed effects.

Our findings indicate that greater exposure to elevated food inflation during third trimester of pregnancy is associated with worse birth outcomes. On average, a one-percentage-point increase in food inflation reduces birthweight by 0.3 grams. The effect is larger (about 0.8 grams) among younger mothers (defined as those under 20 years old), among mothers who are both Black and less educated, the effect is about 0.6 grams. To provide intuition for the magnitude of these estimates, consider that a 7-percentage-point increase in food inflation, an event observed in more than five percent of our sample, would reduce birthweight by approximately 5.6 grams among younger mothers (0.8×7). Higher food inflation has an even smaller impact on birth outcomes among more educated mothers. The timing of exposure is crucial for interpreting the estimated effects. Across nearly all specifications, food inflation is statistically significant when exposure occurs during the third trimester, in some cases during the second trimester, and rarely statistically significant during the first trimester of pregnancy.

Consistent with the vulnerability hierarchy we document, children in both cohabiting households (including consensual unions) and legal marriages show better outcomes than those raised by single mothers. For example, while being born into a two-parent household increases

birthweight by 19 grams on average, the corresponding estimate for legal marriage rises to 24 grams. Among less educated mothers, the advantage increases from 27 grams in two-parent households to 40 grams under legal marriage. A similar pattern emerges for young mothers (23 versus 44 grams) and for mothers who are both Black and less educated (27 versus 41 grams). The presence of a second adult in the household is associated with improved birth outcomes, formal marriage is linked to an additional advantage, potentially reflecting greater economic stability, commitment, or access to resources. The two-parent advantage is heterogeneous with respect to maternal education: mothers with lower educational attainment derive greater benefits from cohabitation, whereas the effect is substantially attenuated among more educated mothers.

Our findings carry some policy implications for mitigating intergenerational poverty. First, we identify which groups of mothers are most vulnerable to food price inflation. Second, we pinpoint the stage of pregnancy during which fetal development is most sensitive to such shocks. Third, we document a tight correlation between birth outcomes in the Brazilian Semiarid and family structure, showing that the gains from two-parent households are largest precisely among historically disadvantaged groups, suggesting that co-resident paternal support during pregnancy may act as a partial buffer against socioeconomic vulnerability, pointing to the importance of household-level support programs for at-risk mothers

Our paper builds upon prior research that explores the relationship between in-utero factors and children's health outcomes. For instance, Currie et al. (2022) and Koppensteiner and Manacorda (2016) have explored the impact of exposure to violence on birth outcomes, while Koppensteiner and Menezes (2023) have focused on the effect of dengue infection. We are closely aligned with literature examining the association between economic fluctuations and birth outcomes. Bozzoli and Quintana-Domeque (2014) and Cruces et al. (2012) investigated the impact of the Argentina Economic Crisis on birth outcomes. Additionally, Bhalotra (2010) examined how economic fluctuations affect infant mortality in India. However, our study is mainly connected to research that links price fluctuations with birth outcomes. Notable contributions in this area include Grace et al. (2014), Kidane and Woldemichael (2020), Woldemichael et al. (2022), and Headey and Ruel (2023).

Using data from the Ethiopian Demographic and Health Survey, Woldemichael et al. (2022) examine the impact of food inflation during pregnancy and infancy on child health, identifying

a causal link between rising food prices and an increased risk of stunting. Although their primary outcome variable (stunting) differs from ours, which focuses on birthweight, their empirical strategy closely resembles ours and yields qualitatively similar results. Similarly, Kidane and Woldemichael (2020) use Ethiopian data to assess the impact of early-life exposure to food price inflation, finding a causal relationship between rising food prices and increased child mortality rates. Headey and Ruel (2023), using a multi-country dataset from the Demographic and Health Surveys Program covering 1.27 million preschool children across 44 developing countries, analyze the relationship between food price shocks and child wasting and stunting. Their results reveal a significant association between rising food prices and increased risks of both outcomes.

Grace et al. (2014), using data from the Kenyan Demographic and Health Survey, investigate the effect of maize price inflation on the incidence of low birthweight. Contrary to expectations, they do not find a straightforward negative relationship. Instead, the coefficient for maize prices in their regression is positive, which they attribute to the fact that some households are net maize producers and benefit from higher prices. However, their interaction term between maize prices and local food production indicates that for households more dependent on market-purchased maize, the effect is indeed negative. While their outcome variable is closely related to ours (newborn weight) their focus is on the binary incidence of low birthweight rather than treating birthweight as a continuous variable.

Our contribution extends beyond Grace et al. (2014) in several dimensions. First, we treat birthweight as a continuous outcome rather than a binary indicator, allowing for more precise estimation of effect sizes. Second, our administrative data spanning 23 years provides substantially greater statistical power than survey-based approaches. Third, we document heterogeneous effects across maternal education, age, and race, groups that bear disproportionate exposure to food price shocks. Finally, and most distinctively, we introduce family structure as an additional layer of vulnerability, showing that the two-parent advantage in birth outcomes is largest precisely among historically disadvantaged mothers.

Following this introduction, the subsequent section of this paper provides background information, presenting stylized facts about the Brazilian semiarid region. Following this, we delve into an explanation of our data sources and empirical strategy. The subsequent section

presents our results, which are compared with existing literature. Finally, we conclude the paper in the last section, offering insights and policy implications from our findings.

2. The Brazilian Semi-arid: Socioeconomic Features and Food Consumption

The Brazilian Semi-arid region spans approximately 18.2% of the national territory, covering 1.563 million square kilometers and including 1,477 municipalities as of the 2024 delineation (Brasil, 2024). This region is home to around 27 million people, accounting for 13% of Brazil's population (IBGE, 2024). The Semi-arid faces significant water shortages due to intense sunlight, high temperatures, and a concentrated rainfall pattern lasting three to four months (Silva et al., 2010). This water scarcity contributes to lower development levels in the region (see Figure 1).

As of 2010, the latest year for which municipal human development index data is available in 2025, no city in the Brazilian Semi-arid region had a higher human development index than the Brazilian national average (IBGE, 2010). Figure 1 illustrates this gap through municipal Human Development Index (HDI) data from 1970 and 2010. The pattern is clear: semi-arid municipalities maintained lower HDI scores than the national average in both periods. This forty-year persistence demonstrates the region's chronic socioeconomic disadvantage, with some improvement in relative standing over time but a persistent ranking gap.

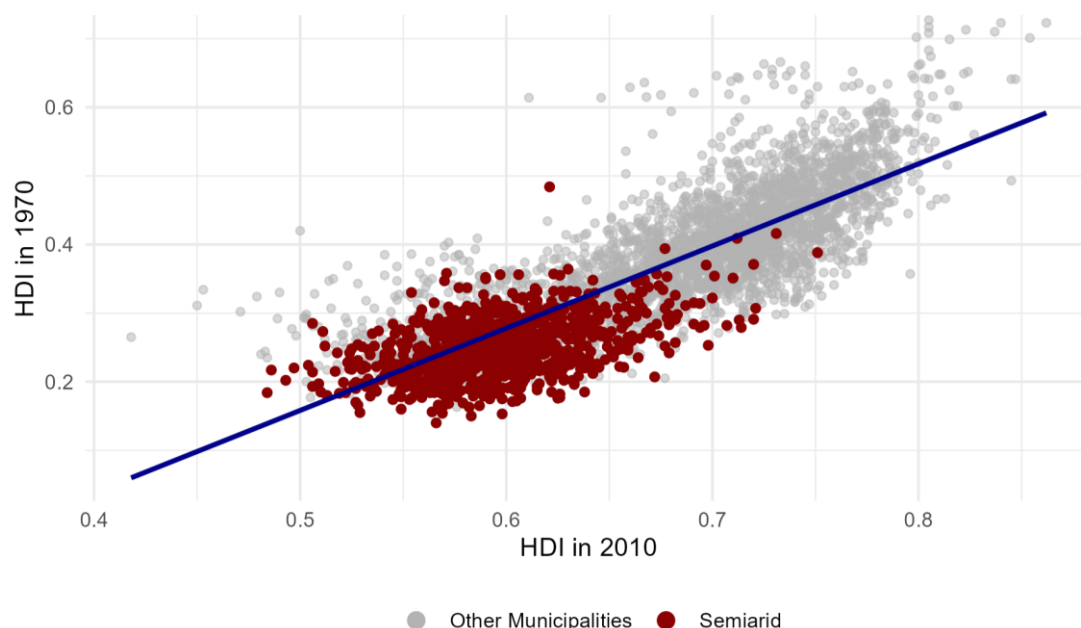


Figure 1- Human Development Index (HDI) in Brazilian Municipalities in 1970 and 2010. The scatter plot compares municipal HDI levels in 1970 (vertical axis) and 2010 (horizontal axis). Dark red dots represent municipalities in the semiarid region; gray dots denote all other municipalities. The blue trend line illustrates the overall relationship between HDI in 1970 and 2010. We are using 2010 HDI data because, as of 2025, it remains the most recent official measure available in Brazil. Although the 2020 Census was expected to provide updated figures, its publication has been postponed by the federal government and is now anticipated for release in 2025. Source: Brasil (2010).

Regarding poverty and urbanization, the semiarid zone exhibits higher poverty rates and a greater concentration of population in rural areas than the rest of the country. Figure 2a illustrates the number of individuals receiving Bolsa Família, a primary poverty alleviation initiative in Brazil, relative to the total population in municipalities within the semiarid zone and the rest of the country. To qualify for the program, individuals must fall below the poverty line in Brazil. Thus, Figure 2a highlights that, on average, cities in the semiarid zone have a higher relative number of people experiencing poverty. In Figure 2b, we analyzed the proportion of households in rural areas for each city in Brazil using data from the Brazilian Census of 2010. The analysis shows that the semiarid zone has a higher proportion of people living in rural areas.

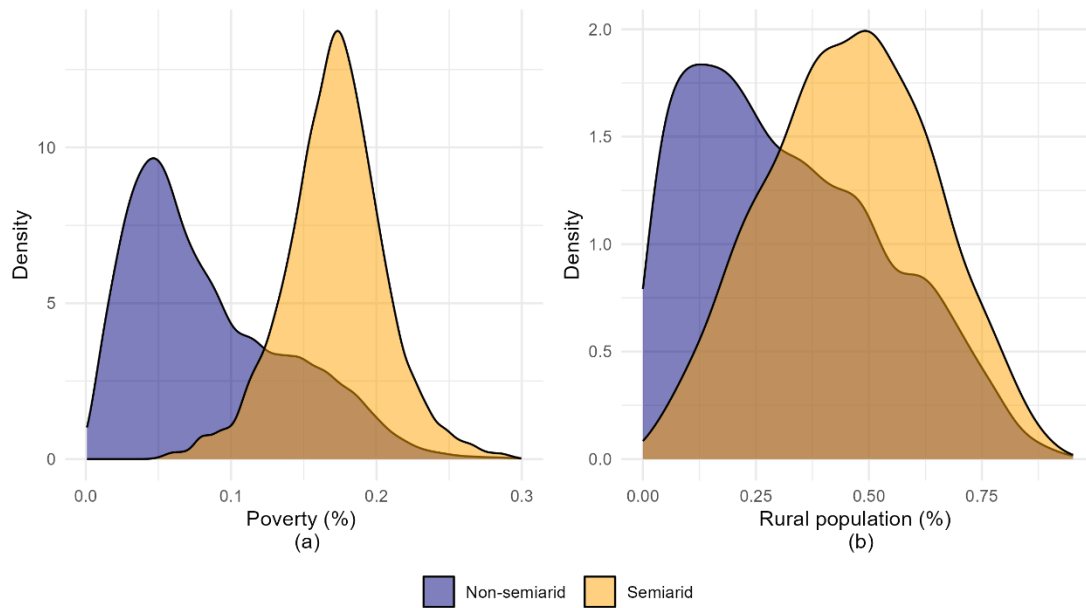


Figure 2- Proxy for Poverty and Rural Household Distribution. Figure (a) depicts the share of adults receiving Bolsa Família benefits relative to the total population (2013-2021). This data reflects the proportion of adults benefiting from Bolsa Família in each municipality in Brazil, recorded monthly from 2013 to 2021 (Source: Brasil, 2024b). Actual poverty rates may be higher because we calculate poverty rates based on the number of adults receiving Bolsa Família divided by the total population; poverty faced by children is ignored. Figure (b) illustrates the proportion of households classified as rural relative to total households in 2010 (Source: Brasil, 2010). Rural areas include peri-urban zones with agricultural activity, low-density neighborhoods, and communities likely to lack basic urban infrastructure (e.g., paved roads, piped water, sanitation). These areas often face marginalization similar to traditional rural poverty, despite lying within municipal boundaries. The metric thus captures both geographic and socioeconomic dimensions of disadvantage. We are using 2010 data for the proportion of rural households because, as of 2025, it remains the most recent official data available.

We could continue comparing numerous social indicators in the semiarid region with those of the rest of Brazil. However, the consistent pattern of poorer outcomes in the semiarid persists across a wide range of indicators and, regrettably, over time.

Regarding food consumption, we lack city-level food consumption surveys in Brazil with sufficient granularity to provide insights specific to the semiarid zone. Our best proxy for this analysis is to examine food consumption patterns in the Northeast Region of Brazil, where most municipalities in the Brazilian Semiarid are located (see Figure A1, in the Appendix). In this region, individuals in the very low, low, and lower-middle-income classes typically allocate over 20% of their budget to food expenditures (refer to Figure 3). This reflects a well-known

pattern whereby more vulnerable households devote a larger share of their income to food.

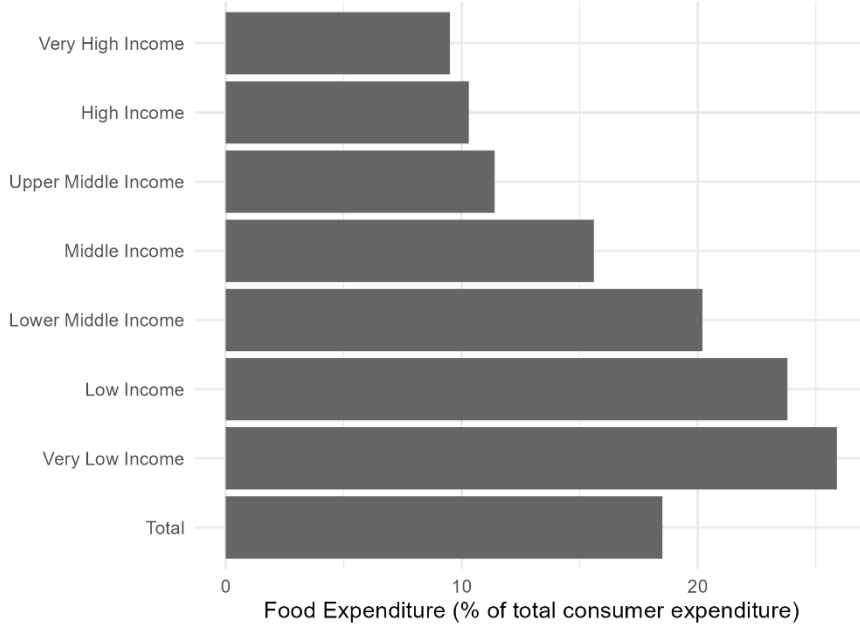


Figure 3 - Share of Expenditure on Food as a Percentage of Total Consumer Expenditure in the Northeast Region of Brazil, based on data from the Consumer Expenditure Survey (POF 2017/2018). Income classes are defined as follows: Very Low Income (up to 1,908 BRL), Low Income (1,908 to 2,862 BRL), Lower Middle Income (2,862 to 5,724 BRL), Middle Income (5,724 to 9,540 BRL), Upper Middle Income (9,540 to 14,310 BRL), High Income (14,310 to 23,850 BRL), Very High Income (over 23,850 BRL). In 2018, the exchange rate was 1 USD = 4 BRL.

3. Data and Empirical Strategy

We quantify the impact of food inflation on birthweight using Equation 1.

$$Y_{ict} = \sum_{k=1}^3 \beta_k \cdot FI_{ct}^k + \gamma \cdot TP_i + X_i' \cdot \delta + \mu_c + \tau_t + \varepsilon_{ict} \quad (1)$$

where Y_{ict} denotes the birthweight (in grams) of child i , born in municipality c in year t . FI_{ct}^k represents the food inflation experienced in municipality c during trimester k of pregnancy. The coefficients β_k capture the effect of exposure to food price inflation during each trimester of gestation. TP_i is an indicator equal to one if the mother lives in a two-parent household (married or in a consensual union), and zero otherwise. X_i' is a vector of maternal and child characteristics obtained from Brazil's Live Birth Information System (SINASC), including maternal education, maternal age, race, and the child's sex. The model includes municipality fixed effects μ_c and year fixed effects τ_t , which control for time-invariant differences across

municipalities and common shocks affecting all births in a given year. ε_{ict} is an idiosyncratic error term. Because we observe the exact date of each birth, we match food inflation exposure to the precise timing of pregnancy. For simplicity, however, the regression notation abstracts from these timing details.

We can estimate separate regressions for each trimester, as in Almond et al. (2011) and Deschênes et al. (2019), or include trimester-specific dummies in a single regression, which is also methodologically valid and adopted in studies such as Andalón et al. (2016) and Koppensteiner and Menezes (2024). In this paper, we follow the first approach.

Brazil’s vital statistics systems – the Sistema de Informações sobre Mortalidade (SIM) and the Sistema de Informações sobre Nascidos Vivos (SINASC) – are key sources for birth outcomes research in the country. SINASC is based on the Declaração de Nascido Vivo (DNV), a nationally standardized live birth certificate managed by the Ministry of Health. The DNV follows a tripartite distribution: one copy is retained by the health facility and sent to municipal health authorities, another is given to the family for civil registration, and the third is used to issue the official birth certificate. SINASC provides critical data on neonatal health, delivery conditions, gestational stages, and maternal health, enabling epidemiological profiling of newborns, analysis of reproductive trends, and the development of healthcare policies. These data are widely used to construct health and demographic indicators in Brazil (Saldanha, 2023)³.

From SINASC, we use birthweight (measured in grams) as well as indicators for low birthweight (less than 2,500 grams) and very low birthweight (less than 1,500 grams). Each observation in the dataset corresponds to a birth and includes information on gestational age, maternal age, and other maternal characteristics, such as the city of residence. We combined these data with food inflation measures, allowing us to capture both the full profile of each birth and the levels of food inflation experienced during different trimesters of pregnancy.

We measure food inflation (FI) using monthly changes in Brazil’s official Consumer Price Index (IPCA), focusing exclusively on its food and beverages component. To construct our local food inflation measure, we follow three steps. First, we compute the average monthly

³ For researchers working with Brazilian health data, Raphael Saldanha represents an essential resource. His contributions include both a comprehensive guide to SUS health databases (Saldanha, 2023) and co-authorship of the microdatasus R package (Saldanha et al., 2019).

inflation rate across the food and beverages items. Second, we calculate the average inflation rate for each trimester of pregnancy as follows:

- Food inflation in the last trimester of pregnancy: the average value of our food inflation index over the three months preceding the child's birth;
- Food inflation in the second trimester of pregnancy: the average value of our food inflation index between the sixth and fourth months preceding the child's birth;
- Food inflation in the first trimester of pregnancy: the average value of our food inflation index between the ninth and seventh months preceding the child's birth.

Therefore, we have computed a food inflation variable for the three metropolitan regions surveyed in the IPCA. Figure 4 shows that food inflation exceeded fifteen percent in 2003 and that, in the upper decile, it averages approximately 7.4 percent.

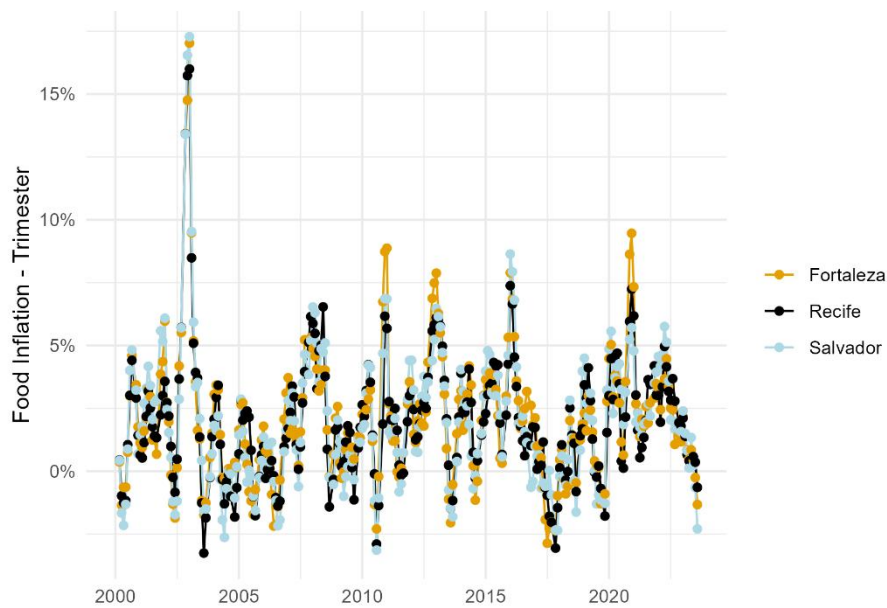


Figure 4 – Average Trimester Food Price Inflation in Three Brazilian Metropolitan Regions (Fortaleza, Recife, and Salvador). Data from January 2000 to January 2023, sourced by the National Consumer Price Index (IPCA) from the Brazilian Institute of Geography and Statistics (IBGE).

Food inflation varies regionally but is not explicitly measured for each municipality in the semiarid zone. Instead, drawing from regional economics concepts, we posit that the food inflation experienced by city (i) can be approximated by the food inflation in the nearest metropolitan region. In short, the proxy for food inflation data encountered by each child (i)

during each in-utero trimester corresponds to the food inflation of the nearest metropolitan region to the mother's residence, determined by Euclidean distance. This approach results in three distinct influence zones within the Brazilian Semi-arid, where the food inflation of each municipality depends on its proximity to either Salvador, Recife, or Fortaleza. Please refer to Figure 5 for a map illustrating these influence zones.

Equation 1 enables us to estimate a distinct effect for each trimester of pregnancy ($k=1, 2, 3$). Our parameter of interest, β_k , quantifies the impact of fetal exposure to food inflation during a specific trimester of pregnancy.

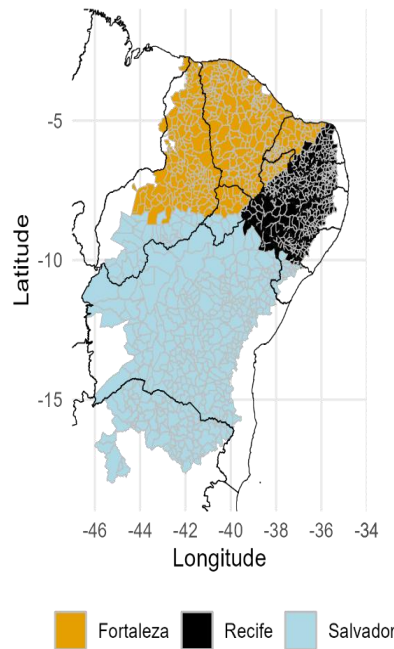


Figure 5 – Influence zones regarding food prices in each city of the Brazilian Semi-arid. The figure displays the spatial nearest neighbor match. Cities shaded in light blue are those closest to the Salvador Metropolitan Region, cities shaded in black are closest to Recife, and cities shaded in dark orange are closest to Fortaleza.

To examine heterogeneity within the sample, we classify mothers according to their education level. Mothers with more than seven years of schooling are defined as more educated, while those with seven or fewer years are classified as less educated. In Brazil, seven years of schooling corresponds to completing the first stage of elementary education and part of the second stage, but falls short of the full nine years required to complete elementary school. In addition to this binary classification, some specifications use the full set of disaggregated

maternal schooling categories available in the data. The final rows of Table 1 report maternal occupation and week of gestation. Although these variables are relevant controls, they are included only in selected specifications because their inclusion substantially reduces the sample size.

We refer to these multiple dimensions of disadvantage as constituting a vulnerability hierarchy. At the top of this hierarchy sit mothers who are simultaneously single, less educated, Black, and young — groups facing compounding disadvantages in income, social support, access to nutritional information, and historical structural barriers. The ordering along each dimension is consistent throughout our results: single mothers are more vulnerable than partnered mothers; among single mothers, the less educated face larger risks; Black mothers and younger mothers confront additional exposures. This hierarchy predicts, and our results confirm, that the negative impact of food inflation on birth outcomes is systematically amplified at lower rungs of the hierarchy, and that the benefit of two-parent cohabitation is largest precisely where other resources are scarcest.

Table 1 – Summary Statistics

<i>Variable</i>	<i>Mean</i>	<i>Std</i>	<i>Max</i>	<i>Min</i>	<i>Observations</i>
<i>Birthweight (g)</i>	3220.18	555.88	9999	0	8,661,586
<i>Low Birthweight</i>	0.08	0.26	1	0	8,661,586
<i>Very Low Birthweight</i>	0.01	0.11	1	0	8,661,586
<i>Food Inflation (3rd)</i>	2.06	3.33	19.62	-5.03	8,584,581
<i>Food Inflation (2nd)</i>	2.14	3.40	19.62	-5.03	8,466,843
<i>Food Inflation (1st)</i>	2.03	3.39	19.62	-5.03	8,353,303
<i>Two-Parent Household</i>	0.51	0.50	1	0	8,696,697
<i>Legally Married</i>	0.35	0.48	1	0	8,585,966
<i>Low Maternal Schooling</i>	0.48	0.50	1	0	8,479,271
<i>Newborn male</i>	0.51	0.50	1	0	8,696,697
<i>Mother Age</i>	25.25	6.58	65	10	8,696,697
<i>City</i>	-	-	-	-	8,696,697
<i>Year</i>	2010.71	6.76	2023	2000	8,696,697
<i>Week of Gestation</i>	38.00	0.00	46	3	4,107,178
<i>Mother Occupation</i>	-	-	-	-	6,659,244

Notes: Authors' elaboration using data from the Brazilian Ministry of Health, specifically the Sistema de Informação sobre Nascidos Vivos (SINASC), and the Brazilian Institute of Geography and Statistics (IBGE). Birthweight refers to the newborn's weight at birth measured in grams. Low birthweight (lbw) is a dummy variable equal to one if birthweight is below 2,500 grams, and zero otherwise. Very low birthweight (vlbw) equals one if birthweight is below 1,500 grams. Two-parent is a dummy variable equal to one when the mother reports living with a partner (either legally married or in a consensual union) and zero otherwise. Legally married is a dummy variable equal to one when the mother reports being legally married. Low maternal schooling is a dummy equal to one when the mother has fewer than eight years of schooling, corresponding to SINASC categories "none," "1–3 years," or "4–7 years," and zero otherwise. The city variable represents the official seven-digit municipality

codes from IBGE and reflects the hierarchical coding structure across municipalities in the Brazilian Semiarid. Newborn male is a dummy equal one if the newborn is male, zero otherwise.

In the regressions, maternal age is also grouped into categories to capture nonlinear age effects across the life cycle. We construct a dummy variable for Black mothers. Following the methodology of the Brazilian Institute of Geography and Statistics (IBGE), the category “Black” corresponds to mothers self-identifying as ‘preta’ (Black) or ‘parda’ (Brown).

We anticipated that local food inflation would have a more pronounced impact on less educated mothers due to their lower income levels and reduced access to food in terms of quality and quantity. Additionally, less educated mothers may have limited access to information about how malnutrition can affect the health of a child in utero. Moreover, in line with existing literature that highlights the influence of marital status on birth outcomes (Shan et al., 2011), we also explore the effects of marital status on the impact of food inflation.

4. Effects of Food Inflation and Family Structure on Newborn Health

Baseline Results

Our baseline results, presented in Table 2, highlight a consistent association between family structure and birth outcomes. Across all specifications, children born to mothers living in two-parent households exhibit higher birthweight than those born to single mothers. In the full sample (Column 1), being born into a two-parent household is associated with an increase of 19 grams in birthweight. The magnitude of this advantage varies across socioeconomic groups. Among less educated mothers (Column 2), the estimated two-parent advantage rises to 27 grams, while among more educated mothers (Column 3) it falls to 13 grams. These differences suggest that the presence of a second adult in the household plays a larger role among more vulnerable mothers. A similar pattern appears across demographic groups. Among Black mothers (Column 4), the estimated advantage is 20 grams, while among young mothers (Column 5) it increases to 23 grams. The advantage is also substantial among mothers who are both young and Black (Column 6), where the estimated increase in birthweight reaches 23 grams. The largest effects are observed among mothers who are Black and less educated (Column 7), for whom the estimated birthweight increase reaches 27 grams. In contrast, among

white and more educated mothers (Column 8), the estimated advantage is considerably smaller, approximately 11 grams.

The birthweight advantage associated with two-parent households is systematically larger among mothers facing greater socioeconomic vulnerability. The interaction specification (Column 9) reinforces this interpretation. The coefficient on the interaction between two-parent households and low maternal schooling is positive and statistically significant, indicating that the two-parent advantage is amplified among less educated mothers. The baseline two-parent effect in this specification is 13 grams, while the interaction term adds an additional 12 grams for less educated mothers, implying a total two-parent advantage of approximately 25 grams in this group.

Regarding food inflation, across nearly all specifications in Table 2, food inflation during the third trimester of pregnancy is associated with statistically significant reductions in birthweight. In the full sample (Column 1), a one-percentage-point increase in food inflation during the third trimester reduces birthweight by 0.3 grams. The magnitude of this effect is somewhat larger among more vulnerable groups. Among less educated mothers (Column 2), the estimated reduction reaches 0.4 grams, while among Black mothers (Column 4) the reduction is also approximately 0.4 grams. The strongest effects appear among young mothers (Column 5) and young Black mothers (Column 6), where a one-percentage-point increase in food inflation during the third trimester reduces birthweight by 0.7 grams and 0.8 grams, respectively.

Table 2 – Two parent advantage and food inflation on birthweight

	Full Sample	Less Educated	More Educated	Black	Young	Young and Black	Black and Less Educated	White and More Educated	Interaction
Two Parent	18.95*** (1.12)	26.59*** (1.09)	13.09*** (1.44)	19.49*** (1.24)	22.79*** (1.69)	22.80*** (1.84)	26.91*** (1.21)	11.21*** (1.75)	13.14*** (1.43)
Food Inflation (3 rd)	-0.30*** (0.09)	-0.36*** (0.13)	-0.26* (0.14)	-0.39*** (0.10)	-0.72*** (0.17)	-0.83*** (0.19)	-0.54*** (0.14)	-0.30 (0.27)	-0.31*** (0.09)
Food Inflation (2 nd)	0.02 (0.11)	-0.20 (0.13)	0.26* (0.16)	0.00 (0.12)	0.07 (0.18)	0.11 (0.21)	-0.27* (0.15)	0.13 (0.29)	0.02 (0.11)
Food Inflation (1 st)	0.07 (0.10)	-0.06 (0.14)	0.24* (0.13)	0.07 (0.11)	0.15 (0.18)	0.18 (0.21)	-0.13 (0.16)	0.16 (0.25)	0.08 (0.10)
Low Schooling									-15.67*** (1.35)
Two Parent × Low Schooling									12.44*** (1.40)
Num.Obs	7676344	3600767	4075577	6117384	1669931	1348755	2863259	821452	7676344
Std.Err.	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id
<i>Fixed Effects:</i>									
City	yes	yes	yes	Yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	Yes	yes	yes	yes	yes	yes
Newborn sex	yes	yes	yes	Yes	yes	yes	yes	yes	yes
Mother age	yes	yes	yes	Yes			yes	yes	yes
Race	yes	yes	yes		yes				yes
Schooling	yes			Yes	yes	yes			

Notes: Standard errors clustered at the municipality level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Birthweight is measured in grams. Two-parent equals one if the mother is married or in a civil union, and zero otherwise.

The estimated effects of food inflation during the second trimester and the first trimester are generally smaller and less consistent. In most specifications these coefficients are statistically indistinguishable from zero. Some isolated results emerge -- for example, a small negative effect for less educated mothers in the second trimester (Column 2) and a negative coefficient among Black and less educated mothers (Column 7) -- but these estimates lack the systematic pattern observed for third-trimester exposure. Similarly, some positive coefficients appear among more educated mothers in Columns 3 and 8, but these estimates likely reflect statistical noise rather than meaningful behavioral responses. Overall, the evidence suggests that late

pregnancy is the period during which fetal development is most sensitive to food price shocks, whereas inflation occurring earlier in pregnancy has a much weaker and less consistent association with birth outcomes.

The concentration of effects in the third trimester is consistent with the developmental biology of fetal growth. From approximately gestational week 28 until delivery, the fetus undergoes its most rapid period of weight accumulation, gaining roughly 70 percent of final birthweight through fat deposition and muscle development, it requires sustained and substantial caloric and protein intake (Jain et al., 2022). Maternal nutritional stress during this window, whether through reductions in dietary quality or quantity induced by higher food prices, directly constrains the substrates available for fetal growth and translates into lower birthweight at delivery. By contrast, earlier trimesters are primarily characterized by organogenesis and structural organ formation. While it is critical, it is also comparatively resilient to short-term nutritional fluctuations. This biological timeline aligns precisely with our empirical finding that third-trimester food inflation is the dominant predictor of reduced birthweight, while first- and second-trimester exposure shows weaker and less systematic associations.

Marriage versus Two-Parent Households

To examine whether the estimated two-parent advantage reflects the institutional dimension of marriage rather than simply the presence of two adults in the household, we re-estimate the baseline specifications by restricting the definition of partnership to legally married mothers. In the SINASC data, this corresponds to mothers whose marital status is recorded as married, excluding those reported as living in a consensual union. Thus, while the two-parent dummy in our main analysis is broader and includes consensual unions, the definition based on legal marriage is more restricted. The results are presented in Table 3.

The estimates remain strongly consistent with the baseline findings and show an even larger advantage associated with formal marriage. In the full sample (Column 1), being born to a legally married mother is associated with an increase of approximately 26 grams in birthweight, compared with the 19-gram advantage observed for two-parent households in Table 2. As before, the magnitude of this effect varies across socioeconomic groups. Among less educated mothers (Column 2), the estimated marriage advantage reaches about 40 grams.

A similar pattern emerges across demographic groups. The estimated birthweight advantage associated with marriage is about 27 grams among Black mothers (Column 4) and exceeds 40 grams among young mothers (Column 5) and among mothers who are both young and Black (Column 6). Overall, the comparison between Tables 2 and 3 suggests that the estimated two-parent advantage becomes even larger when consensual unions are excluded and partnership is defined strictly as formal marriage, reinforcing the idea that the institutional aspects of marriage may provide additional stability or resources that benefit early-life health outcomes.

Table 3 - Formal marriage advantage and food inflation on birthweight

	Full Sample	Less Educated	More Educated	Black	Young	Young and Black	Black and Less Educated	White and More Educated	Interaction
Legally married	23.90*** (0.92)	39.66*** (1.05)	13.13*** (0.98)	25.44*** (0.85)	44.13*** (1.44)	45.23*** (1.66)	40.69*** (1.14)	10.74*** (1.88)	14.46*** (1.09)
Food Inflation (3 rd)	-0.27*** (0.09)	-0.29** (0.13)	-0.26* (0.14)	-0.36*** (0.10)	-0.66*** (0.17)	-0.78*** (0.19)	-0.46*** (0.15)	-0.28 (0.27)	-0.27*** (0.09)
Food Inflation (2 nd)	0.05 (0.11)	-0.15 (0.13)	0.26* (0.16)	0.03 (0.12)	0.10 (0.18)	0.15 (0.21)	-0.22 (0.15)	0.14 (0.29)	0.04 (0.11)
Food Inflation (1 st)	0.07 (0.10)	-0.05 (0.14)	0.24* (0.13)	0.07 (0.11)	0.14 (0.18)	0.16 (0.21)	-0.12 (0.16)	0.17 (0.25)	0.08 (0.10)
Low Schooling									-15.07*** (1.12)
Two Parent × Low Schooling									22.18*** (1.43)
Num.Obs	7647926	3587575	4060351	6094387	1663718	1343683	2852568	818532	7647926
Std.Err.	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id
<i>Fixed Effects:</i>									
City	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes	yes	yes	yes
Newborn sex	yes	yes	yes	yes	yes	yes	yes	yes	yes
Mother age	yes	yes	yes	yes			yes	yes	yes
Race	yes	yes	yes		yes				yes
Schoolin g	yes			yes	yes	yes			

Notes: Standard errors clustered at the municipality level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Birthweight is measured in grams. Legally married is a dummy equal to one if the mother's marital status is married and zero otherwise. Observations corresponding to civil unions are excluded from this table.

We summarize these differences in Figure 6, which plots side by side the estimated effects of being born to a legally married mother and to a mother living in a two-parent household. In both cases, children born in these family structures exhibit higher birthweight relative to those born to single mothers. However, the advantage associated with legal marriage is consistently larger than the one associated with two-parent households more broadly. This difference is particularly pronounced among less educated mothers, suggesting that the institutional aspects of marriage may provide additional stability or resources that translate into improved birth outcomes.

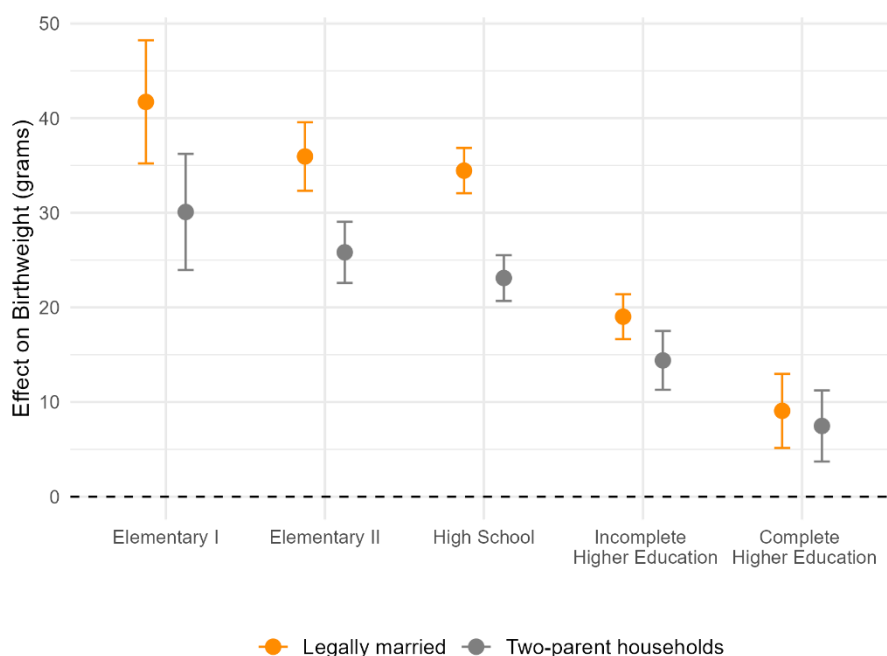


Figure 6 – Effects of Family Structure on Birthweight by Maternal Education. The figure displays estimated coefficients and 95% confidence intervals from regressions including municipality and year fixed effects, as well as controls for newborn sex, maternal race, maternal age, and maternal education. Gray markers represent estimates for children born in two-parent households, while dark orange markers represent estimates for children born to legally married mothers. The horizontal axis reports the mother’s level of education.

COVID Outbreak

The COVID-19 pandemic generated disruptions in both healthcare systems and food markets. Hospitals faced unusually high demand, while food prices had sharp increases driven by supply

chain disruptions and changes in consumption patterns. For robustness, we re-estimate our baseline specifications excluding births occurring during the pandemic period. The results are reported in Table 4.

The results excluding the COVID-19 period are presented in Table 4 and closely mirror the baseline findings. The estimated two-parent advantage remains stable and statistically significant across specifications, with children born in two-parent households exhibiting higher birthweight than those born to single mothers, particularly among more socioeconomically vulnerable groups. Similarly, the effects of food inflation follow the same pattern observed in the baseline estimates. Food inflation during the third trimester continues to be associated with reductions in birthweight, while inflation during the first and second trimesters shows weaker and less consistent associations. Overall, excluding the COVID-19 period does not alter the main conclusions.

Table 4 – Two parent advantage and food inflation on birthweight (excluding Covid period)

	Full Sample	Less Educate d	More Educate d	Black	Young	Young and Black	Black and Less Educate d	White and More Educate d	Interactio n
Two- parent	20.43** *	27.30** *	14.34** *	21.26** *	24.20** *	24.42** *	27.73** *	11.50** *	14.40***
	(1.05)	(1.09)	(1.38)	(1.14)	(1.58)	(1.74)	(1.19)	(1.69)	(1.38)
Food Inflation (3 rd)	-0.42*** (0.10)	-0.42*** (0.13)	-0.45*** (0.15)	-0.55*** (0.11)	-0.80*** (0.17)	-0.93*** (0.19)	-0.62*** (0.15)	-0.35 (0.29)	-0.43*** (0.10)
Food Inflation (2 nd)	-0.22* (0.12)	-0.31** (0.13)	-0.11 (0.19)	-0.28** (0.13)	-0.19 (0.19)	-0.20 (0.22)	-0.41*** (0.15)	-0.01 (0.29)	-0.23** (0.11)
Food Inflation (1 st)	-0.15 (0.10)	-0.13 (0.14)	-0.16 (0.14)	-0.21* (0.12)	-0.07 (0.19)	-0.12 (0.22)	-0.23 (0.16)	0.02 (0.28)	-0.15 (0.10)
Low Schooling									-15.03*** (1.33)
Two Parent × Low Schooling									11.77*** (1.42)
Num.Obs	6604456	3389114	3215342	5182109	1501311	1197759	2670146	703379	6604456
Std.Err.	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id
<i>Fixed Effects:</i>									
City	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes	yes	yes	yes
Newborn sex	yes	yes	yes	yes	yes	yes	yes	yes	yes
Mother age	yes	yes	yes	yes			yes	yes	yes
Race	yes	yes	yes		yes				yes
Schoolin g	yes			yes	yes	yes			

Notes: Standard errors clustered at the municipality level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Birthweight is measured in grams. Legally married is a dummy equal to one if the mother's marital status is married and zero otherwise. Observations corresponding to civil unions are excluded from this table.

Incidence of low birthweight

Up to this point, our focus has been on birthweight as a continuous outcome. However, one potential concern is that changes in average birthweight may be driven by shifts at the upper tail of the distribution – for instance, by a reduction in the number of very large babies – rather

than by improvements among infants at risk. To address this issue, we replicate our analysis using the incidence of low birthweight (LBW) as the outcome variable (equals one if the newborn weighs less than 2,500 grams and zero otherwise). The results are presented in Table 5.

Across all specifications, births occurring in two-parent households exhibit significantly lower odds of low birthweight. In the full sample, the coefficient of -0.091 implies approximately 9 percent lower odds of LBW relative to births to single mothers, corresponding to a reduction of roughly 0.6 percentage points from a baseline prevalence of 7.5 percent. As in the birthweight regressions, the two-parent advantage is larger among socioeconomically disadvantaged mothers. The coefficient is -0.11 among less-educated mothers compared to -0.07 among those with higher education. Similarly, the association is particularly pronounced among Black mothers with low education (-0.11). The interaction specification confirms this pattern: the two-parent advantage is significantly larger among mothers with low schooling.

Regarding food inflation, higher food inflation during the third trimester is consistently associated with an increased probability of low birthweight, although its effect on average birthweight appears smaller in magnitude. Food inflation during the second and first trimesters does not display a clear pattern in relation to LBW, and in some specifications the estimated coefficients even suggest the opposite of what would be expected -- that higher food inflation is associated with a lower probability of low birthweight. Taken together, these results suggest that exposure to food price shocks late in pregnancy may modestly increase birth risks.

Table 5 – Logistic Regression: Probability of Low Birthweight

	Full Sample	Less Educated	More Educated	Black	Young	Young and Black	Black and Less Educated	White and More Educated	Interactio n
Two-parent households	- 0.0908** *	- 0.1149** *	- 0.0732** *	- 0.0903** *	- 0.0858** *	- 0.0830** *	- 0.1106** *	- 0.0661** *	- 0.0804***
	(0.0081)	(0.0080)	(0.0098)	(0.0088)	(0.0116)	(0.0121)	(0.0086)	(0.0130)	(0.0095)
Food Inflation (3 rd)	0.0019** *	0.0026** *	0.0010	0.0014**	0.0042** *	0.0036** *	0.0019**	0.0019	0.0019***
	(0.0006)	(0.0009)	(0.0008)	(0.0006)	(0.0011)	(0.0012)	(0.0009)	(0.0019)	(0.0006)
Food Inflation (2 nd)	- 0.0022** *	-0.0017**	- 0.0028** *	- 0.0025** *	-0.0028**	-0.0032**	-0.0019**	-0.0017	- 0.0022***
	(0.0007)	(0.0009)	(0.0010)	(0.0008)	(0.0012)	(0.0013)	(0.0009)	(0.0023)	(0.0007)
Food Inflation (1 st)	-0.0001	-0.0007	0.0004	-0.0002	-0.0014	-0.0021	-0.0008	0.0001	-0.0001
	(0.0006)	(0.0009)	(0.0009)	(0.0007)	(0.0011)	(0.0013)	(0.0010)	(0.0020)	(0.0006)
Low Schooling									0.0621*** (0.0077)
Two Parent × Low Schooling									- 0.0227*** (0.0077)
Num.Obs	7676344	3600767	4075577	6117384	1669931	1348709	2863259	821136	7676344
Std.Err.	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id	by: id
<i>Fixed Effects:</i>									
City	yes	yes	yes	yes	yes	yes	yes	yes	yes
Year	yes	yes	yes	yes	yes	yes	yes	yes	yes
Newborn sex	yes	yes	yes	yes	yes	yes	yes	yes	yes
Mother age	yes	yes	yes	yes			yes	yes	yes
Race	yes	yes	yes		yes				yes
Schooling	yes			yes	yes	yes			

Notes: The dependent variable equals one if the newborn weighs less than 2,500 grams and zero otherwise. Coefficients are estimated using logistic regression models. Standard errors clustered at the municipality level are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Two parent households is a dummy equal to one if the mother is legally married at the time of birth and zero otherwise. Observations corresponding to civil unions are excluded.

Comparing our results with the literature

Several studies demonstrate that economic and social interventions during pregnancy can significantly influence birth outcomes. Our findings reveal a particularly pronounced association between family structure and newborn health. Infants born to mothers in two-parent households weigh between 19 and 27 grams more at birth than those born to single mothers, with even larger advantages when partnership is defined strictly as legal marriage. These magnitudes are comparable to the effects documented for cash transfer programs in Uruguay, which increased birthweight by 23 to 30 grams (Amarante et al., 2016), and for access to household water cisterns in Brazil, which raised birthweight by approximately 46 grams (Da Mata et al., 2023). Our estimates thus fall within the range of improvements generated by targeted social interventions during pregnancy.

The estimated effects of food inflation, while smaller in magnitude, remain consistent with the literature on prenatal shocks. Exposure to high food inflation during the third trimester reduces birthweight by approximately 8 grams in the most adverse scenario. This is comparable to findings by Camacho (2008), who reports that an earthquake in Colombia reduced birthweight by roughly 8 grams, and by Rocha and Soares (2015), who find that a one-standard-deviation change in rainfall patterns in Brazil's semiarid region is associated with a shift of approximately 1.6 grams in birthweight. Although modest in absolute terms, these estimates fall within the range of impacts documented for transitory shocks affecting maternal conditions during pregnancy.

Placebo Test: Future Food Inflation

A key identification concern is whether our food inflation measure captures a genuine prenatal causal effect or instead proxies for broader economic conditions that affect birth outcomes through unobserved channels. To assess this, we implement a placebo test using future food inflation. We construct the cumulative food inflation rate during months 1–3 after the child's birth (f_1), and the rate during months 4–6 after birth (f_2), following the same construction as t_1 and t_2 but using forward rather than backward lags of the monthly price series. Since post-birth food inflation cannot causally affect birthweight, any statistically significant association would

indicate that our main estimates capture spurious economic correlations rather than the prenatal channel.

We estimate three specifications: (A) the baseline model including the in-utero trimester variables t_1 , t_2 , and t_3 (replicating Table 2, Column 1); (B) a placebo model that replaces the in-utero variables with the post-birth measures f_1 and f_2 ; and (C) a joint model that includes all five variables simultaneously, allowing the data to distinguish whether birthweight responds to past or future food prices. The results are reported in Table 6.

Table 6 – Falsification Test: Replacing In-Utero Trimester Variables with Post-Birth Inflation

	(A) Baseline	(B) Falsification	(C) Joint
Two-parent	18.950*** (1.124)	19.824*** (1.065)	19.139*** (1.119)
Food Inf. 3rd trimester [in-utero]	-0.302*** (0.095)		-0.304*** (0.099)
Food Inf. 2nd trimester [in-utero]	0.022 (0.108)		0.024 (0.107)
Food Inf. 1st trimester [in-utero]	0.072 (0.098)		0.066 (0.100)
Food Inf. Months 1–3 post-birth [Placebo]		0.107 (0.092)	0.015 (0.093)
Food Inf. Months 4–6 post-birth [Placebo]		0.129 (0.094)	-0.004 (0.096)
Number Observations	7,676,344	7,902,178	7,607,365

Notes: Standard errors clustered at the municipality level in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Birthweight is measured in grams. Two-parent equals one if the mother is married or in a civil union. Column (A) replicates the baseline specification of Table 2 (full sample). Column (B) replaces in-utero trimester variables with post-birth placebo measures f_1 and f_2 . Column (C) includes both in-utero and post-birth measures simultaneously. Controls in all columns: newborn sex, maternal race, maternal age group, maternal education, municipality fixed effects, year fixed effects.

These results support a causal interpretation of the third-trimester food inflation effect and provide no evidence that our estimates reflect spurious economic trends rather than genuine prenatal exposure.

5. Final Remarks

This paper examines two distinct but complementary determinants of birth outcomes in the Brazilian Semi-arid: food inflation during pregnancy and family structure. Using administrative records on more than seven million births between 2000 and 2023, we find that exposure to higher food inflation during the third trimester of pregnancy reduces birthweight, with effects concentrated among the most vulnerable mothers. A one-percentage-point increase in food inflation reduces birthweight by 0.3 grams on average, but the effect reaches 0.8 grams among young mothers and 0.6 grams among Black mothers with low education. Regarding family structure, children born into two-parent households are, on average, 19 grams heavier at birth than those born to single mothers, with the advantage rising to 27 grams among less educated mothers and to over 40 grams when partnership is defined strictly as legal marriage. These results control for a set of observables and they are robust to a wide range of specifications and hold when the COVID-19 period is excluded, confirming that the estimated effects are not driven by pandemic-related disruptions to health services or food markets.

Our findings consistently point black, less educated and young mothers as the most vulnerable. It reflects deep-rooted socioeconomic inequalities that compound across multiple dimensions of prenatal risk. Identifying these groups is, in our view, among the most practically relevant contributions of this paper, as it provides a concrete basis for targeting prenatal support programs toward those who need them most.

The two-parent advantage in birthweight raises an important question about policy. One might be tempted to interpret our results as an argument for promoting marriage or discouraging single parenthood. We caution strongly against such a reading. Family structure is not a policy lever in any straightforward sense. Societies have changed, partnership decisions are deeply personal, and the diversity of family arrangements is a feature of modern life that policy should accommodate rather than resist. What our results show, however, is that single mothers – and particularly those with low education – face a structural disadvantage that translates directly into worse birth outcomes for their children. The appropriate policy response is not to make it harder to be a single mother, but to make it less costly: through targeted prenatal nutrition programs, income support, and expanded access to prenatal care for this group. Closing the gap

between single and partnered mothers in the Brazilian Semiarid likely requires interventions designed specifically with this group in mind, rather than general-purpose transfers.

Our results tell a story about the origins of inequality that begins before birth. A child born to a young, Black, less educated, single mother in the Brazilian Semiarid during a period of high food inflation already enters the world at a measurable biological disadvantage relative to a child born under more favorable circumstances. This is not merely a statistical regularity – it is the earliest observable manifestation of a process of intergenerational disadvantage transmission.

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Appendix

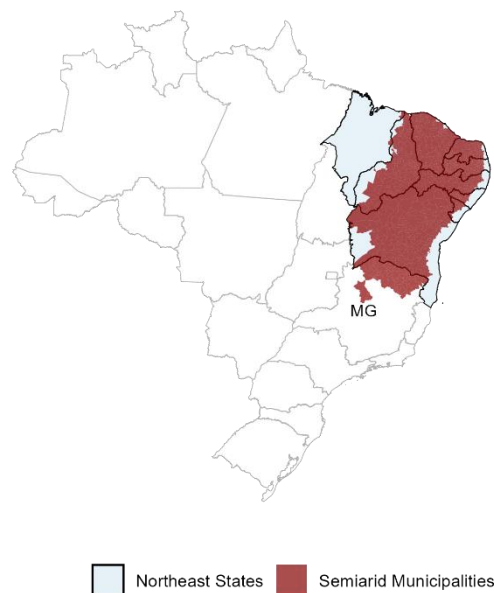


Figure A1 – Delimitation of the Northeast Region and Semi-arid Municipalities in Brazil. The Northeast Region includes all states shown in light blue. Semi-arid municipalities, based on the 2017 official definition used in this study, are highlighted in dark red. Note that some semi-arid municipalities are also located in northern Minas Gerais (labeled “MG” on the map), outside the Northeast Region.